

Working section prototype, 10 September 2026. This text tests the additional body of the Hodge part. The numerical results recalled below belong to the preceding part of the proposed paper.

1 Hodge conservation after one stabilization

The numerical argument selected whole quantum primary factors whose contributions vanish on surface centers. We now retain the odd Hodge representation of those factors. The new point is to restrict the quantum parameters while keeping the entire cohomology fiber.

Let $\mathcal{F}$ consist of the smooth complex Picard-rank-one Fano threefolds of index two and degree 1, 2, 3, and of index one and genus 2, 3, 4, 5, 6, 8.

Theorem 1 (Rational Hodge conservation). *For $X, Y \in \mathcal{F}$, a birational map $X \times \mathbf{P}^1 \dashrightarrow Y \times \mathbf{P}^1$ implies an isomorphism*

$$H^3(X, \mathbf{Q}) \cong H^3(Y, \mathbf{Q})$$

of rational Hodge structures.

The numerical part supplies four ingredients used here: regular separation of whole primary connection blocks; cyclic persistence in all even bulk directions and rank-two residue transport; the low-dimensional vanishing calculations; and the nine-family endpoint computations and specialized $\mathbf{P}^1$ lemma. We reuse these proofs on the bases specified below. The fixed-base comparison and the final recovery of a rational Hodge structure are proved in this section. No assertion of the numerical classification requires this theorem.

1.1 A common group and the retained fiber

For a polarizable rational Hodge structure, the Hodge group is the rational algebraic group generated by the circle action that acts on $H^{p,q}$ as λ^{p-q} . Tate twists do not change this action. Iritani recalls its universal form in [1, Section 3]: the category of polarizable rational Hodge structures modulo Tate twists has a rational fiber functor, and its tensor automorphisms form the universal Hodge group.

Fix the finite collection of varieties and centers in a projective weak factorization of the proposed birational map, together with the auxiliary surface models used below. Use one reductive rational quotient G of this universal group through which all their cohomology representations factor. The image on each cohomology space is its Hodge group. Thus the choices for different varieties are compatible. Tate suspensions in the blowup formula act trivially on these representations.

For each variety T , restrict the bulk parameter to

$$H^*(T, \mathbf{Q})^G \otimes_{\mathbf{Q}} \mathbf{C}, \quad \text{while the fiber remains } H^*(T, \mathbf{Q}) \otimes_{\mathbf{Q}} \mathbf{C}.$$

The fixed parameters are even: the element -1 of the Hodge circle acts by cohomological parity. The unit and algebraic divisor classes are fixed. No assertion that every Hodge class is algebraic is needed. Novikov parameters, z and the coefficient ramification parameters have trivial G -action and even parity.

The quantum product is equivariant under simultaneous action on parameters and fibers [1, Lemma 7]. On this fixed base its Euler operator therefore commutes with G . Its whole primary projectors commute with G as well. All ranks below are full even fiber ranks, never dimensions of G -invariant subspaces.

After a splitting extension of the generic coefficient field, record the odd part of each primary factor by its multiplicities in the irreducible representations of $G_{\mathbf{C}}$. These give a class in the representation group $R(G_{\mathbf{C}})$, a free abelian group on simple representations. Further algebraically closed coefficient extension does not change these multiplicities.

1.2 Faithfulness on the fixed base

The comparison must recover the intrinsic generic center connection, even when its raw Novikov map identifies curve classes. We check faithfulness on the fixed base directly.

Lemma 2 (Fixed-base comparison). *For a blowup $\text{Bl}_Z T \rightarrow T$ with smooth connected center of codimension $c \geq 2$, the actual comparison on Hodge-fixed bulk bases preserves whole primary factors, their full G -representations and their original z -lattices. Each center occurrence is a faithful extension of its intrinsic reduced fixed-base connection. Eligible rank-two factors retain their canonical modified residue discriminant.*

Proof. Iritani's blowup isomorphism intertwines the quantum connections over the specified graded completed $\mathbf{C}[z]$ -modules [2, Theorem 5.18 and Section 5.8.2]. Its coordinate maps and full-fiber isomorphism are equivariant, and their initial shifts are fixed [1, Proposition 8 and Lemma 10]. The joint coordinate map and its inverse restrict to inverse maps of fixed loci. In the external direct sum the fixed coordinates of the ambient copy and each center copy are independent; in particular their unit and divisor coordinates survive.

Write d for an effective integral numerical curve class of Z . The reduced source retains the combined generator $X_d = Q_Z^d \exp(\sigma^{(2)} \cdot d)$, with no separate source divisor variables. If u denotes fixed higher-degree coordinates and v the unit coordinate, its ring is

$$R_Z^G = \mathbf{C}[[X_d, u]]_{\text{gr}}[v].$$

Here graded completion retains only finitely many total homogeneous degrees in each element, and unit dependence is polynomial. At a fixed curve class the coefficient is consequently polynomial in u, v : every u has strictly negative degree. These are precisely the reduced-domain conventions of the numerical comparison proof.

For the j th occurrence, with independent fixed divisor coordinate $s_j^{(2)}$ and fixed initial shift ς_j° , the map is

$$X_d \mapsto Q^{i_* d} q^{-c_1(N_{Z/T}) \cdot d / (c-1)} \exp((\varsigma_j^{\circ, (2)} + s_j^{(2)}) \cdot d).$$

The other coordinates undergo the corresponding fixed shifts. Existence of this reduced substitution uses the string and divisor equations; it does not authorize translation of unrestricted formal power series.

To prove injection, take the least nonzero degree with respect to the restriction of an ambient ample divisor. Bounded numerical effective-degree slices are finite. At this least degree, positive-degree corrections to the initial shifts cannot contribute; translation of the polynomial coefficients by their degree-zero shifts is injective. Group terms with the same untagged target monomial. A kernel element would give a nonzero finite relation

$$\sum_d a_d \exp\left(\sum_k (D_k \cdot d) x_k\right) = 0,$$

where the a_d are independent of the divisor coordinates x_k . Choose integral algebraic divisors D_k separating numerical curve classes. Their coordinates remain available on the fixed base. Restrict to an integral line in these coordinates separating the finitely many

exponent vectors. Successive derivatives at zero give an invertible Vandermonde matrix, forcing all $a_d = 0$. This is a contradiction.

The original comparison and its inverse are regular at $z = 0$ in the coefficientwise realization of the graded completion. Leading-term intertwining therefore carries N and $\text{im } N$ to their counterparts, and hence carries the canonical modification

$$E^\sharp = \{s \in E : s \bmod z \in \text{im } N\}$$

to the corresponding modification. In modified frames the regular inverse still exists, so the residues are conjugate. This applies also to resonant rank-two factors. Finally, independent occurrence-unit shifts separate their spectra generically: the resultant of two scalar-translated characteristic polynomials is a nonzero polynomial in the shift difference. Thus distinct occurrences do not merge in the comparison. $\square$

The same argument applies to the ambient copy and componentwise to a disconnected center. Each operation uses its own comparison field; the resulting equalities of representation multiplicities telescope without composing different Laurent expansions along the factorization.

1.3 Selection and vanishing

Let E run over the whole primary factors of the full connection on the fixed base. Retain E if either

$$\text{rk } E^{\text{ev}} = 2, \quad N_E^2 = 0 \neq N_E, \quad \delta^\sharp(E) \neq 0, \quad \text{or} \quad \text{rk } E^{\text{ev}} = 3.$$

Here N_E is the centered leading operator on the even part, and $\delta^\sharp$ is the canonical rank-two residue discriminant. Define

$$\mathcal{H}_G(T) = \sum_{E \text{ retained}} [E^{\text{odd}}] \in R(G_{\mathbf{C}}).$$

There is no factor of one half. Selection precedes extraction of the odd representation. The sum has no scalar eigenvalue or discriminant labels; permuting splitting branches does not change it. We do not require each individually labelled factor to descend to $\mathbf{Q}$.

Proposition 3 (Operation and vanishing). *The selected class satisfies*

$$\mathcal{H}_G(\text{Bl}_Z T) = \mathcal{H}_G(T) + (c - 1)\mathcal{H}_G(Z).$$

It is zero for smooth projective varieties of dimension at most two and is unchanged under birational maps of smooth projective fourfolds.

Proof. The fixed-base comparison preserves the selection conditions and the odd representations, with the $c - 1$ center Tate suspensions forgotten by G . Generic separation of occurrences makes their contributions add.

The low-dimensional calculations from Part I apply to these bases as follows. A point has only an even-rank-one factor. A curve has simple even factors, zero centered operator, or a rank-two factor with discriminant zero. For a minimal surface with nef canonical class, the centered Euler operator strictly raises ordinary degree, also on the fixed base. Its only primary factor has full even rank $b_2 + 2$. The only possible selected rank is three, when $b_2 = 1$. In that case a nonzero holomorphic one-form α would give a nonzero class $[i\alpha \wedge \bar{\alpha}]$ of square zero, impossible in the one-dimensional second cohomology generated by an ample class. Thus H^1 and H^3 vanish, and its odd contribution is zero.

On $C \times \mathbf{P}^1$ with $g(C) \geq 1$, the full-even-bulk calculation in Part I has two even-rank-two factors, each with centered operator $(2 - 2g(C))h$. For genus one it is identically zero; for larger genus the modified residue discriminant is zero. All even classes here are Tate, so this is also the fixed-base calculation. The horizontal Novikov parameter remains an independent spectator. For $C = \mathbf{P}^1$ the generic factors are simple. Every ruled surface is birational to such a product; surface factorization uses point centers only, whose contributions have already been shown to vanish. The $\mathbf{P}^2$ calculation and point blowups complete the surface classification argument. No fourfold invariance was used to prove these vanishings.

Projective weak factorization of fourfolds now uses only smooth projective centers of dimension at most two. Every correction is zero, proving the last assertion. $\square$

1.4 Endpoint recovery and the proof of the theorem

We first explain why continuation preserves the information just retained. On a connected fixed-base germ an equivariant projector has constant representation multiplicities. Indeed, write the full fiber as a sum $\bigoplus_i V_i \otimes M_i$ of irreducible $G_{\mathbf{C}}$ -representations and multiplicity spaces. The projector acts by the identity on V_i and an idempotent on M_i . Its image has locally constant rank, hence constant rank on the germ. This argument concerns a separated continued projector; it does not continue a decomposition through a collision of eigenvalues.

For $X \in \mathcal{F}$, all even cohomology is Tate and $H^{\text{odd}}(X) = H^3(X)$. The nine endpoint calculations in Part I give one repeated even primary factor: rank two with nonzero discriminant in index-two degrees 1, 2, 3 and index-one genera 6, 8, or rank three in index-one genera 2, 3, 4, 5. All complements have even rank one.

An even-rank-one factor of a Frobenius superalgebra has no odd part. Writing its even unit as e , odd elements satisfy $ab = ce$; supercommutativity gives $(ab)^2 = 0$, hence $c = 0$. Nondegeneracy of the odd Frobenius pairing then forces the odd part to vanish. Consequently the entire $H^3(X)$ lies in the unique retained repeated factor, and

$$\mathcal{H}_G(X) = [H^3(X, \mathbf{Q})_{\mathbf{C}}].$$

At the small product point, the genus-zero product formula identifies the full connection and its original lattice for $X \times \mathbf{P}^1$ with the tensor product [3, formula (1)]. The independent fiber Novikov parameter separates the two $\mathbf{P}^1$ eigenvalues and every shifted X cluster. The specialized $\mathbf{P}^1$ lemma of Part I continues these clusters in all even bulk directions, including mixed ones: cyclicity and even rank persist, and the rank-two discriminant is unchanged. Tensoring with the regularly separated rank-one factor changes the residue only by a scalar. On the fixed base the equivariant-projector argument above preserves the odd multiplicities. The Tate-shifted second copy has the same G -representation. Therefore

$$\mathcal{H}_G(X \times \mathbf{P}^1) = 2[H^3(X, \mathbf{Q})_{\mathbf{C}}].$$

This uses neither a big quantum tensor formula at mixed bulk nor a general projective-bundle comparison theorem.

Proof of Theorem 1. Choose the common group G for a weak factorization between the two products. Operation and vanishing, followed by endpoint recovery, give

$$2[H^3(X, \mathbf{Q})_{\mathbf{C}}] = 2[H^3(Y, \mathbf{Q})_{\mathbf{C}}] \quad \text{in } R(G_{\mathbf{C}}).$$

Semisimplicity makes this group free on simple classes. Cancel two and obtain an isomorphism of the complexified G -representations.

Put $V = H^3(X, \mathbf{Q})$ and $W = H^3(Y, \mathbf{Q})$. Both have pure weight three. On their Hodge summands the circle records $p - q$, and the common weight records $p + q = 3$; together these determine (p, q) . Thus a rational G -equivariant map between V and W is precisely a rational Hodge morphism. Equivalently, no nonzero Tate twist can identify weight-three constituents here.

Finally, equivariant Hom commutes with scalar extension, so

$$\text{Hom}_G(V, W) \otimes_{\mathbf{Q}} \mathbf{C} = \text{Hom}_{G_{\mathbf{C}}}(V_{\mathbf{C}}, W_{\mathbf{C}}).$$

The determinant polynomial on the finite-dimensional rational vector space $\text{Hom}_G(V, W)$ is nonzero because the complexified space contains an isomorphism. Over the infinite field $\mathbf{Q}$, a nonzero polynomial has a rational point where it does not vanish. That point gives the required rational Hodge isomorphism. $\square$

The conclusion retains neither a distinguished integral lattice nor a principal polarization. Those structures are separate inputs in any subsequent Torelli or arithmetic application.

References

- [1] H. Iritani, *Notes on the decomposition theorem for blowups*, arXiv:2604.10028v2, Section 3, Lemmas 7 and 10, Proposition 8.
- [2] H. Iritani, *Quantum cohomology of blowups*, arXiv:2307.13555v3, Remark 2.3, Theorem 5.18 and Section 5.8.2.
- [3] K. Behrend, *The product formula for Gromov–Witten invariants*, arXiv:alg-geom/9710014v1, formula (1).